Journal of Circular Economy (2026) 4:2, No. 17
https://doi.org/10.55845/e3j0xq90

PERSPECTIVE

Rethinking Smart Packaging for the Circular Economy

Justina Karpavice[1*], Promporn Wangwacharakul[2], Malgorzata Fialkowska-Filipek[3]

Received: 30 June 2026 / Accepted: 3 August 2026 / Published: 20 August 2026
© The Author(s) 2026

Abstract

Smart packaging (SP) has emerged as a key enabler of circular economy (CE) strategies, transforming packaging from a passive container into an active data interface. Yet, despite extensive knowledge of SP's technical capabilities, adoption and commercialisation remain limited, and circular actions require active human engagement that data availability alone cannot guarantee. This perspective conceptualizes SP through a socio-technical lens, arguing that its contribution to CE outcomes depends on coordinated interactions among technological capabilities, human actors, and organizational processes across the supply chain. Drawing on Human-Computer Interaction (HCI) and technology acceptance perspectives, we argue that stakeholder engagement with SP is shaped by a range of human, technological, and contextual factors that have received limited attention in CE research. We highlight the need for greater integration of HCI and technology acceptance perspectives into SP research and advocate human-centric SP design as a necessary condition for realizing its CE potential.

Keywords Socio-Technical Systems · Human-Computer Interaction · Technology Acceptance · Stakeholder Engagement · Circular Supply Chain

1. Introduction

As CE strategies extend beyond end-of-life waste management towards system-level resource optimization, packaging is increasingly recognised as a strategic interface that links local operational activities with broader sustainability goals, facilitating circularity, resilience, and sustainable value creation (Ciravegna, 2025). SP has emerged as a key enabler of circular supply chains through the integration of intelligent, active, and connected technologies, including RFID tags, QR codes, and sensors (Wangwacharakul et al., 2026). These technologies transform packaging from a passive container into an active data interface capable of continuously collecting, storing, and transmitting information throughout the product lifecycle (Fiałkowska-Filipek et al., 2026; Lydekaityte & Tambo, 2020). As a result, SP enhances end-to-end traceability, facilitates real-time data exchange, and supports information sharing among supply chain stakeholders, thereby improving visibility and coordination across value chains (Wangwacharakul et al., 2026; Grisi et al., 2026). These capabilities create new opportunities to support CE strategies by enabling data-driven decision-making, regulatory compliance, and more effective real-time monitoring of material and product flows (Fiałkowska-Filipek et al., 2026; Wahab et al., 2025; Mehra et al., 2026). Consequently, SP can contribute to reduced food waste and spoilage, enhanced food safety, and more efficient resource utilisation (Grisi et al., 2026; Wahab et al., 2025; Mehra et al., 2026).

Despite the growing recognition of SP as an enabler of circular supply chains, existing research remains predominantly technology-centric, focusing on the development, functionalities, and advancement of

* Corresponding author: justina@btech.au.dk
[1] Department of Business Development and Technology, Aarhus University, Herning, Denmark.
[2] Department of Management and Engineering, Linkoping University, Linkoping, Sweden.
[3] Department of Management Systems and Organizational Development, Faculty of Management, Wrocław University of Science and Technology, Wrocław, Poland.

technologies and digital infrastructures, associated with SP. This includes RFID tags, sensors, indicators, QR codes, nanomaterials, blockchain, artificial intelligence, and digital traceability systems (Rajan & Wani, 2025; Balbinot-Alfaro et al., 2026; Mehra et al., 2026; Grisi et al., 2026; Siracusa & Lotti, 2019). While these studies have substantially advanced understanding of SP's technical capabilities, considerably less attention has been given to the mechanisms through which these capabilities are translated into circular actions in practice. This gap reflects a broader paradigmatic assumption in supply chain management research — that outcomes follow from technological capability rather than from the conditions that shape human engagement with it (Gualandris et al., 2024). This may help explain why, despite the knowledge of SP technologies and their potential benefits, broader adoption and commercialisation remain limited in real-world settings (Butuc et al., 2025). Accordingly, we suggest that the remaining challenge may lie less in technological capability than in how people engage with it. Information generated by SP has no intrinsic circular value, i.e. its contribution to CE depends on whether that information is recognised, interpreted, and translated into action by multiple stakeholders throughout the product lifecycle. While SP can generate, store, and communicate information, circular actions such as scanning, returning, sorting, refilling, and recycling require active human engagement (Wangwacharakul et al., 2026; Lydekaityte & Tambo, 2020).

The aim of this perspective piece is to reconceptualize SP through a socio-technical lens and argue that its effectiveness in generating CE outcomes depends on how multiple stakeholders engage with, interpret, and act upon the information it provides. In doing so, the paper contributes to the literature by shifting the design question from what SP is capable of transmitting to whether and how manufacturers, logistics operators, retailers, consumers, and recyclers can and will act on that information. While circular supply chain research typically considers these actors at the organizational level, engagement with SP-enabled information ultimately occurs through individual users — such as production operators, warehouse workers, consumers, and recycling operators — acting within these organizations. Therefore, to understand how digital capabilities translate into circular outcomes, we highlight the need to complement the macro-level supply chain view with HCI and technology acceptance perspectives, identifying the factors that shape individual user engagement with SP-enabled information.

The remainder of the paper is organised as follows. Section 2 develops the paper's theoretical position by conceptualising SP through socio-technical systems theory, positioning it within a multi-actor supply chain context, and introducing HCI and technology acceptance perspectives. Section 3 discusses the implications of this perspective for future research and practice and concludes the paper.

2. Theoretical Positions

2.1. SP from a Socio-Technical Systems Perspective

Packaging has long been understood as more than a material artefact. Carli Lorenzini & Olsson (2019) describe it as "a socio-scientific discipline which operates in society to ensure the delivery of goods to the ultimate consumer", emphasising that packaging extends beyond its material form and is embedded within broader social systems. Similarly, Ratajczyk (2025) conceptualises packaging as a Socio-Technical System (STS) of material and form (artifact), design and service (process), and actors (producers, consumers, recyclers), oriented towards creating value for people and the planet. SP extends this systemic character further. Through applied design research, Ciravegna (2025) demonstrates that it functions as a dynamic node engaging simultaneously with the contained products, stakeholders, and broader infrastructures through which it circulates. Within this context, SP is most productively understood not as a standalone technological artifact but as a socio-technical interface embedded within a broader STS, in which CE outcomes emerge from the interaction between social and technical elements rather than from technological capability alone (Maguire, 2014; Baxter & Sommerville, 2011). STS theory identifies four defining characteristics: a collective operational task; interdependent social and technical sub-systems; openness to environmental change; and a degree of incompleteness that demands ongoing adaptation (Eason, 2010; Maguire, 2014). As Maguire (2014) demonstrates across a range of organisational contexts, even technically adequate systems fail when the interface between users and technology is designed without attention to socio-technical factors - a finding with direct implications for SP. Applying this framework, SP can be understood as comprising interdependent technical sub-systems enabling data capture, storage and sharing (e.g. RFID, QR codes, and sensors) and social

sub-systems comprising the people, roles, practices, and decisions through which stakeholders (e.g. manufacturers, logistics operators, retailers, consumers, and recyclers) engage with SP-enabled information, with both sub-systems contributing to the collective operational task of enabling circular material flows. Crucially, the open and unfinished character of STS means that SP should support different stakeholders with different roles, interaction needs, and levels of discretion across the product lifecycle - from production and logistics through to use and end-of-life management. CE outcomes, in this view, depend on the joint optimisation of the technical and social elements of the system rather than solely on technological functionality. Consequently, HCI and technology acceptance provide essential analytical lenses for understanding how different stakeholders engage with SP-enabled information across the product lifecycle (Maguire, 2014).

2.2. SP within a Multi-Actor Supply Chain Context

Supply chains are inherently characterised by continuous interactions among diverse stakeholders, interconnected logistical processes, and complex flows of products and information (Appolloni et al., 2023). Within these networks, packaging occupies a unique position, serving as the central interface among manufacturers, logistics operators, retailers, consumers, and recyclers throughout the product lifecycle (Lydekaityte, 2019). SP acts as a mediator of interactions across the supply chain, from production and logistics to product use and end-of-life management (Ciravegna, 2025). From the operational point of view, these interactions differ across stakeholders: manufacturers generate and encode product information, logistics operators rely on packaging for identification and tracking, retailers support inventory management and consumer communication, consumers access information to inform purchasing, use, and end-of-life decisions, while recyclers rely on packaging information to enable material identification and resource recovery (Wangwacharakul et al., 2026; Lydekaityte & Tambo, 2020). However, existing SP research has predominantly focused on end consumers (e.g. Shi et al., 2026; Henrichs et al., 2025; Endara et al., 2023; Just & Goddard, 2023; Young et al. 2020), with comparatively less attention given to other supply chain stakeholders, thereby treating the supply chain as a relatively uniform information environment rather than a system of multiple actors with different roles, capabilities, and engagement conditions (Butuc et al., 2025). Consequently, SP should be understood as a multi-user interaction system whose contribution to CE outcomes depends on how well it supports the distinct interaction needs of stakeholders across the product lifecycle.

2.3. The Human Dimension of SP: HCI and Technology Acceptance Perspectives

Once SP is viewed as part of a broader STS, a fundamental question emerges: what shapes stakeholder interaction with the information that SP provides? Two streams of research - HCI and technology acceptance - offer the most direct analytical lens on this question. HCI concerns the design and evaluation of interactive systems in relation to human use, examining how interface characteristics, task context, and user cognition jointly determine whether a system produces its intended effects (Coiera, 2003). The Technology Acceptance Model (TAM), originally proposed by Davis (1989), identifies two constructs as the primary determinants of whether users adopt and act through a technological system: *perceived usefulness* (the degree to which a person believes the system will enhance their performance), and *perceived ease of use* (the degree to which use is expected to be free of effort). These constructs are not merely predictors of adoption; they are design criteria. A system that is functionally capable but perceived as difficult to use or of unclear benefit will not produce the behaviours it is designed to support, regardless of its technical sophistication. Applied to SP, this means that circular actions, e.g. scanning, returning, sorting, refilling, depend not on whether RFID tags or QR codes are present, but on whether stakeholders find interacting with them worthwhile and manageable.

Recent empirical research confirms this logic in circular and pro-environmental contexts. Shi et al. (2026) demonstrate that perceived ease of use and perceived usefulness both significantly predict consumer attitudes toward carbon footprint label engagement on food packaging — and that these attitudes, mediated by perceived behavioural control, shape actual low-carbon purchasing behaviour. Accordingly, this confirms that information availability alone is insufficient to produce circular actions. Furthermore, since its introduction, TAM has evolved through numerous extensions that incorporate additional behavioural, social, and contextual determinants of technology acceptance, enabling its application across diverse technologies and stakeholder contexts (Venkatesh et al., 2003; Venkatesh & Bala, 2008). Existing behavioural research has shown that

attitudes, perceived behavioural control, subjective norms, and environmental concern influence consumers' willingness to participate in circular packaging systems (Ram et al., 2025). Similarly, consumer awareness of the environmental consequences of packaging and well-designed reusable packaging positively influence reuse behaviour (Corona et al., 2024), while collective social attitudes further shape recycling practices (Ram et al., 2025). Communication through packaging, such as dynamic date labels or integrated displays, can further steer consumer behaviour by reducing household food waste (Henrichs et al., 2025). However, this is not confined to consumers. While consumers often interact directly with the physical packaging, other supply chain actors engage with the broader SP system through digital interfaces that enable access to packaging-enabled information. For logistics operators, effective interaction depends not only on system intelligence but also on human-centred interfaces that support supervisory decision-making within increasingly automated logistics environments (Cimini et al., 2020), where usability, security, and user trust shape operators' willingness and ability to access, interpret, and act upon SP-enabled information (Maktoof & Rabeaa, 2026). Furthermore, despite increasing digitization and automation across supply chains, stakeholders continue to perform manual activities such as transporting, sorting, scanning, storing, picking, and distributing products (Mättig & Kretschmer, 2019). Taken together, these examples suggest that the success of SP is determined less by what the technology is capable of doing than by whether stakeholders are willing and able to interact with it effectively. HCI and technology acceptance perspectives are, therefore, not peripheral to the SP agenda; they are the missing analytical foundation for understanding why CE potential often remains unrealised in practice.

3. Discussion and Conclusion

This perspective piece has argued that SP's contribution to CE outcomes depends not on what the technology transmits, but on whether and how multiple actors engage with, interpret, and act upon the information it provides. Framing SP through an STS lens reveals a structural research gap. The human side of the SP-circularity relationship has been treated as peripheral precisely when it should be central. SP generates information; CE outcomes emerge only when that information is transformed into action by multiple stakeholders. HCI and technology acceptance offer the analytical tools to make that human side both legible and designable.

The most significant implication concerns where research attention needs to go. Behavioural factors shaping consumer participation in circular packaging have begun to receive empirical attention (Shi et al., 2026; Corona et al., 2024; Ram et al., 2025; Young et al., 2020). Considerably less is known about the factors shaping interactions across the broader supply chain - among manufacturers encoding product information, logistics operators acting on condition data, retailers managing inventory through packaging interfaces, and recyclers relying on SP to identify and recover materials. Each actor group operates under different institutional conditions, different cognitive workloads, and different relationships to the packaging system; and whether SP generates circular outcomes depends on whether each group finds interacting with it useful and manageable (Davis, 1989; Maguire, 2014; Butuc et al., 2025).

As a perspective paper, this article is intended to provide a conceptual reframing rather than empirical validation. Consequently, the arguments presented should be viewed as a theoretical foundation rather than as empirically verified relationships. The proposed perspective therefore requires empirical examination across different stakeholder groups, technologies, and supply chain contexts. Building on this foundation, three research directions follow. First, empirical studies should investigate technology acceptance constructs across supply chain roles beyond consumers — e.g. the interaction conditions facing a logistics operator differ fundamentally from those facing a consumer at the point of purchase, and role-specific perceived usefulness and ease of use are necessary preconditions for circular behaviour across the full product lifecycle. Second, HCI design principles need to be applied systematically to SP development, which has been dominated by engineering and materials science framings (Wangwacharakul et al., 2026; Fiałkowska-Filipek et al., 2026), since designing for circular intent means designing for the specific interaction contexts of each supply chain actor. Third, the relationship between SP-enabled information and circular behaviour needs to be examined at the supply chain level, where circular actions are distributed across multiple actors and touchpoints in ways that neither individual-level behavioural research nor logistics-level SP research has yet addressed.

For practice, organisations investing in SP for CE purposes should evaluate systems not only on data-generation capabilities but on whether stakeholder groups can and will interact with them as the circular model

requires. Perceived usefulness, ease of use, trust, security, awareness, and similar are design outcomes, not fixed technological properties — and supply chain actors who do not find SP interaction worthwhile or manageable will not undertake circular actions regardless of the system's technical sophistication. Ultimately, the challenge is not to make SP smarter. It is to make it more human-centric.

Author contributions Justina Karpavice, Promporn Wangwacharakul and Malgorzata Fialkowska-Filipek contributed to the conceptualization of the study. Justina Karpavice led the writing of the manuscript with the support of Promporn Wangwacharakul and Malgorzata Fialkowska-Filipek. All authors contributed to the critical revision of the manuscript, approved the final version and agreed to its submission for publication.

Funding The authors declare that no funding was received for this research.

Data availability No new data were created or analyzed in this study.

Declarations

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The images or other third-party material in this article are included in the article's Creative Commons License, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons License and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this license, visit http://creativecommons.org/licenses/by/4.0/.

References

Appolloni, A., D'Adamo, I., Gastaldi, M., Yazdani, M., & Settembre-Blundo, D. (2023). Reflective backward analysis to assess the operational performance and eco-efficiency of two industrial districts. *International Journal of Productivity and Performance Management*, *72*(6), 1608-1626. https://doi.org/10.1108/ IJPPM-08-2021-0442

Balbinot-Alfaro, E., da Rocha, M., Alfaro, A. T., Romani, V. P., Oliveira, A. K. G., da Silva, T. R. C., & Düsman, E. (2026). Smart packaging for food traceability and supply chain management. *Smart and Intelligent Food Packaging*, 173-191. https://doi.org/10.1016/B978-0-443-24724-8.00005-3

Baxter, G., & Sommerville, I. (2011). Socio-technical systems: From design methods to systems engineering. *Interacting with computers*, *23*(1), 4-17. https://doi.org/10.1016/j.intcom.2010.07.003

Butuc, O., Akkerman, R., & De Leeuw, S. (2025). Smart packaging adoption in fresh fruits and vegetables supply chains. *The International Journal of Logistics Management*, *36*(7), 284-307. https://doi.org/10.1108/IJLM-08-2024-0521

Carli Lorenzini, G., & Olsson, A. (2019). Towards patient-centered packaging design: An industry perspective on processes, functions, and constraints. *Packaging technology and science*, *32*(2), 59-73. https://doi.org/10.1002/pts.2419

Cimini, C., Lagorio, A., Romero, D., Cavalieri, S., & Stahre, J. (2020). Smart logistics and the logistics operator 4.0. *IFAC-PapersOnLine*, *53*(2), 10615-10620. https://doi.org/10.1016/j.ifacol.2020.12.2818

Ciravegna, E. (2025). FuturE-Pack. Designing Smart Packaging for Circular and Sustainable Made in Italy. *DA*, 1-160. https://dx.doi.org/10.30682/9791254776711

Coiera, E. (2003). Interaction design theory. *International journal of medical informatics*, *69*(2-3), 205-222. https://doi.org/10.1016/S1386-5056(02)00106-5

Corona, B., Tunn, V. S., & van den Broek, K. L. (2024). Integrating consumer behaviour into the environmental assessment of circular packaging: a scoping review. *The International Journal of Life Cycle Assessment*, *29*(1), 80-98. https://doi.org/10.1007/s11367-023-02218-1

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly*, *13*(3), 319-340. https://doi.org/10.2307/249008

Eason, K. (2010). Before the internet: The relevance of socio-technical systems theory to emerging forms of virtual organisation. *International Journal of Sociotechnology and Knowledge Development*, *1*(2), 23-32. https://doi.org/10.4018/jskd.200904010

Endara, P., Wiedmann, M., & Adalja, A. (2023). Consumer willingness to pay for shelf life of high-temperature, short-time-pasteurized fluid milk: Implications for smart labeling and food waste reduction. *Journal of Dairy Science*, *106*(9), 5940-5957. https://doi.org/10.3168/jds.2022-22968

Fiałkowska-Filipek, M., Karpavičė, J., & Wangwacharakul, P. (2026). Smart packaging as a digital enabler for circularity in sustainable supply chains. *Cleaner Logistics and Supply Chain*, 100299. https://doi.org/10.1016/j.clscn.2026.100299

Grisi, C. V. B., Araújo, P. L., da Silva, R. D. C. A., da Silva Marques, A., de Almeida, A. F., & de Albuquerque Meireles, B. R. L. (2026). Smart and intelligent packaging benefits for quality control and assurance. *Smart and Intelligent Food Packaging*, 111-133. https://doi.org/10.1016/B978-0-443-24724-8.00007-7

Gualandris, J., Branzei, O., Wilhelm, M., Lazzarini, S., Linnenluecke, M., Hamann, R., ... & Chen, C. M. (2024). Unchaining supply chains: Transformative leaps toward regenerating social-ecological systems. *Journal of Supply Chain Management, 60*(1), 53-67. https://doi.org/10.1111/jscm.12314

Henrichs, E., Senge, J. M., & Krupitzer, C. (2025, January). A Consumer Study on the Acceptance of Digitized Packaging for Food Quality Assessment. In *Proceedings of the 58th Hawaii International Conference on System Sciences*, 5880-5889. https://doi.org/10.5281/zenodo.13384187

Just, D. R., & Goddard, J. M. (2023). Behavioral framing and consumer acceptance of new food technologies: Factors influencing consumer demand for active packaging. *Agribusiness*, *39*(1), 3-27. https://doi.org/10.1002/agr.21778

Lydekaityte, J. (2019). Smart Interactive Packaging as a Cyber-Physical Agent in the Interaction Design Theory: A Novel User Interface. In: Lamas, D., Loizides, F., Nacke, L., Petrie, H., Winckler, M., Zaphiris, P. (eds) Human-Computer Interaction - INTERACT 2019. *Lecture Notes in Computer Science, vol 11746.* Springer, Cham. https://doi.org/10.1007/978-3-030-29381-9_41

Lydekaityte, J., & Tambo, T. (2020). Smart packaging: definitions, models and packaging as an intermediator between digital and physical product management. *The International Review of Retail, Distribution and Consumer Research*, *30*(4), 377-410. https://doi.org/10.1080/09593969.2020.1724555

Maguire, M. (2014). Socio-technical systems and interaction design - 21st century relevance. *Applied Ergonomics*, *45*(2), 162-170. https://doi.org/10.1016/j.apergo.2013.05.011

Maktoof, M. A. J., & Rabeaa, M. M. (2026, March). PLC-Based Automated Sorting System in Smart Packaging Industry. In *Proceedings of the International Conference on Applied Innovations in IT* (ICAIIT), *14*(1), 1435-1441. https://doi.org/10.25673/123749

Mättig, B., & Kretschmer, V. (2019). Smart packaging in intralogistics: an evaluation study of human-technology interaction in applying new collaboration technologies. In *Proceedings of the 52nd Hawaii International Conference on System Sciences*, 739-748. https://doi.org/10.24251/HICSS.2019.091

Mehra, K., Sarkar, N., Deepak, Bisht, P., & Kumkum. (2026). Revolutionizing Intelligent Food Packaging: Nanomaterials, AI-ML, and Their Seamless Integration. *Food Reviews International*, 1-33. https://doi.org/10.1080/87559129.2026.2631781

Ram, N., Ahmad, P., Toth-Peter, A., Torres de Oliveira, R., & Acharyulu, G. V. R. K. (2025). Developing a circular economy framework for e-commerce packaging materials: A study on behavioural intentions of online consumers. *Business Strategy and the Environment*, *34*(1), 982-1006. https://doi.org/10.1002/bse.4020

Rajan, S. S., & Wani, K. M. (2025). A review of smart food and packaging technologies: revolutionizing nutrition and sustainability. *Food and Humanity*, *4*, 100593. https://doi.org/10.1016/j.foohum.2025.100593

Ratajczyk, M. (2025). From traditional packaging to packaging 5.0: redefining the role of packaging in Industry 4.0 and Industry 5.0. *Economics and Environment*, *94*(3), 948-948. https://doi.org/10.34659/eis.2025.94.3.948

Siracusa, V., & Lotti, N. (2019). Intelligent packaging to improve shelf life. *Food quality and shelf life*, 261-279. https://doi.org/10.1016/B978-0-12-817190-5.00008-2

Shi, N., Rao, M., Li, J., Wu, Z., & Zhang, J. (2026). Carbon Footprint Rating Information on Food Packaging and Consumer Low-Carbon Purchasing Behavior: An Integrated TPB-TAM Model. *Sustainability*, *18*(10), 4666. https://doi.org/10.3390/su18104666

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *Mis Quarterly*, *27*(3), 425-478. https://doi.org/10.2307/30036540

Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision sciences*, 39(2), 273-315. https://doi.org/10.1111/j.1540-5915.2008.00192.x

Wahab, S. N., Othman, N., Sayuti, N. M., Mohaini, M. L., & Atan, R. (2025). Assessing the growth and trends of smart packaging: A case of food distribution. *Journal of Information Technology Management*, *17*(1), 159-177. https://doi.org/10.22059/jitm.2025.99930

Wangwacharakul, P., Karpavice, J., Fialkowska-Filipek, M., & Chawla, Y. (2026). Smart packaging's role in enhancing logistics capabilities for a circular economy: a systematic literature review. *International Journal of Physical Distribution & Logistics Management*, *56*(5), 520-543. https://doi.org/10.1108/IJPDLM-06-2025-0302

Young, E., Mirosa, M., & Bremer, P. (2020). A systematic review of consumer perceptions of smart packaging technologies for food. *Frontiers in Sustainable Food Systems*, *4*, 63. https://doi.org/10.3389/fsufs.2020.00063